Check for updates

Received: 13 September 2023 | Revised: 2 June 2024 | Accepted: 20 June 2024

DOI: 10.1002/pmrj.13256

ORIGINAL ARTICLE

PM&R

The importance of language in describing concussions: A qualitative analysis

Peter Knowles BA[1,2] | Katherine Schneider BS[2,3] | Amy K. Bugwadia MS[2,4] | Piya Sorcar PhD[2,4,5,6] | Roy D. Pea DPhil[2,6] | Daniel H. Daneshvar MD, PhD[2,7,8,9] | Christine M. Baugh PhD, MPH[10]

[1]Department of Psychology, Stanford University, Stanford, California, USA

[2]TeachAids, Institute for Brain Research and Innovation, Palo Alto, California, USA

[3]Department of Mechanical Engineering, Stanford University, Stanford, California, USA

[4]Stanford University School of Medicine, Stanford, California, USA

[5]Stanford Center for Innovation in Global Health, Stanford University, Stanford, California, USA

[6]Stanford University Graduate School of Education, Stanford University, Stanford, California, USA

[7]Spaulding Rehabilitation Network, Division of Brain Injury Rehabilitation, Spaulding Rehabilitation Hospital, Charlestown, Massachusetts, USA

[8]Harvard Medical School, Boston, Massachusetts, USA

[9]Rehabilitation Outcomes Center at Spaulding, Spaulding Rehabilitation Network, Boston, Massachusetts, USA

[10]Center for Bioethics and Humanities, University of Colorado Anschutz Medical Campus, Aurora, Colorado, USA

Correspondence
Christine M. Baugh, University of Colorado Anschutz Medical Campus, Center for Bioethics and Humanities, 13001 East 17th Place, Aurora, CO 80045, USA.
Email: christine.baugh@cuanschutz.edu

Abstract

Background: Concussions are mild traumatic brain injuries that are often undiagnosed due to difficulties in identifying symptoms. To minimize the negative sequelae associated with undiagnosed concussion, efforts have targeted improving concussion reporting. However, knowing more about concussions does not indicate how likely an athlete is to report their concussion. Alternatively, the attitudes and beliefs of athletes and surrounding stakeholders have shown to be a better indication of whether an athlete intends to report their concussion. Prior research has shown that athletes report concussions less often when the injury is described using language that minimizes their severity, such as when it is referred to as a "ding." This study evaluated whether describing concussions using the word "brain" was associated with individuals' underlying attitudes and beliefs about the injury's severity.

Objective: To measure the relationship between perceived concussion severity and the language used to describe concussions, specifically whether participants used the word "brain" in describing the injury.

Methods: One-on-one semi-structured telephone interviews were conducted, and a cross sectional secondary qualitative analysis was performed to assess participants' perceived concussion severity and their use of the word "brain" to describe concussions.

Design: Cross-sectional secondary qualitative analysis.

Setting: One-on-one semistructured telephone interviews.

Participants: In 2017, 94 individuals involved in high school sports, including athletes, coaches, educators, parents of athletes, and athletic directors were recruited via convenience sampling.

Main Outcome Measures: Respondents' perceived severity of concussions.

Results: Individuals who used a brain phrase to describe concussion also perceived concussions as more severe ($p < .001$). Specifically, those who described concussions with maximum severity had higher odds of using brain phrases than those who described concussions as having minimum (odds ratio [OR] = 0.05, 95% confidence interval [CI] = 0.002–0.299, $p < .001$) or moderate severity (OR = 0.24, 95% CI = 0.086–0.647, $p = .003$), with the most significant relationship found among coaches.

Conclusions: These findings demonstrate the relationship between medical terminology and perceived severity of concussions. This relationship may play a role in concussion reporting behavior for coaches, athletes, and parents. Education programs using similar medical terminology may promote concussion reporting behaviors.

 © 2024 American Academy of Physical Medicine and Rehabilitation.

INTRODUCTION

Concussion is a common injury with an estimated 1.1 to 1.9 million concussions in the United States each year among children younger than 18 years old.[1] It is also estimated that 0.5 million to 1.2 million people with concussions do not receive the proper medical diagnosis.[2–5] Untreated concussions increase the risk of subsequent neurologic and orthopedic injury,[6–8] as well as persisting symptoms after concussion, which can persist for months or years following the brain injury.[8] For youth with a developing brain, untreated concussions raise concerns about potential negative long-term health consequences.[9–13] Sustaining another brain injury following a concussion can lead to the exceptionally rare albeit catastrophic implications of second impact syndrome, including persistent neurologic impairment and death.[8,14,15] These significant burdens associated with undiagnosed concussion highlight the importance of proper injury identification and treatment.[16]

Concussion diagnosis often relies upon athlete disclosure of the injury to a clinician; as a result, research has sought to identify the factors that influence athletes' concussion care-seeking behaviors. Teaching athletes about concussions alone has shown limited increases in concussion reporting.[17,18] However, an athlete's attitudes and beliefs about both concussions and concussion reporting influence concussion care-seeking. Previous work suggests these factors play important roles in facilitating behaviors and therefore carry consequences for health outcomes.[4,19–21] Athletes' attitudes and beliefs are affected by their environment, including coach support and perceived social norms.[22] An athlete may perceive these factors through the language their teammates or coaches use in describing the injury.

Researchers have studied the complex relationship between language and concussions. Findings suggest that the vocabulary used to describe and define concussions has a significant impact on self-reporting.[23–25] This work sought to understand whether the language that athletes and athletic stakeholders use to discuss concussion offers insight into their perceived severity of concussions. We hypothesized that individuals who perceived concussion as more severe would be more likely to use the word "brain" when describing concussions compared to those who perceived concussion as less severe.

METHODS

This research protocol was approved by Stanford University Institutional Review Board, Protocol 40720.

Participants

Participants included high school athletes, coaches, educators, parents, and an athletic director. All participants were personally recruited through convenience sampling by the research team through either personal networks or LinkedIn online searches.

Data collection

Interviews were semistructured and conducted over the phone in July 2017. Verbal consent was obtained from each participant to record the interview. The purpose of the recruitment of these participants for these interviews was to inform the development of "CrashCourse," a concussion education curriculum by the nonprofit organization TeachAids. Later, these interviews were revisited for the secondary analysis presented in this paper. In total, 316 participants were interviewed and 94 were used in analysis. These 94 were selected because interviews were performed by the same interviewer (K.S.) asking the same questions and there were audio recordings to ensure accuracy. Transcripts were never returned to participants for comments or corrections. The interview script was created in discussion with research team members, following a narrative literature review of the topics and education interventions addressed in the concussion literature. The semistructured interview script can be found in Data S1. The interviews lasted an average of 20 minutes and responses were documented with a combination of live note-taking by the interviewer and audio recordings.

Because these interviews were initially conducted as user experience research for the development of "CrashCourse," a concussion education curriculum by the nonprofit organization TeachAids, a post hoc coding scheme was created for the purposes of this research project (Figure 1). The coding criterion was developed by the research team and applied by the interviewer (K.S.) to extract the perceived severity of a concussion and the terminology used to describe concussion. A second coder reviewed audio records for independent verification of coding (P.K.). In order to maintain consistency with the interpretation of the coding criteria, both coders independently flagged quotes for additional discussion with the research team in cases in which the selected quote initially appeared ambiguous and required further deliberation. Twenty-three percent of cases ($n = 22$) were flagged for additional discussion, resulting in changes in classification for 11 cases.

Each participant's perceived severity of concussions was systematically categorized into "minimum," "moderate," or "maximum," based on their responses to two open-ended questions: (1) participants were asked to "rank the severity of the following injuries: stress fracture, concussion, broken arm, torn anterior cruciate ligament (ACL), and sprained ankle," and then provide an explanation of their thought processes; (2) participants were asked, "if an athlete were to get injured and find out they had a concussion, do you think they would choose to go back into the game if presented with the choice?" (Table 1).

19341563, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/pmrj.13256 by Ohio State University Ohio Sta, Wiley Online Library on [01/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

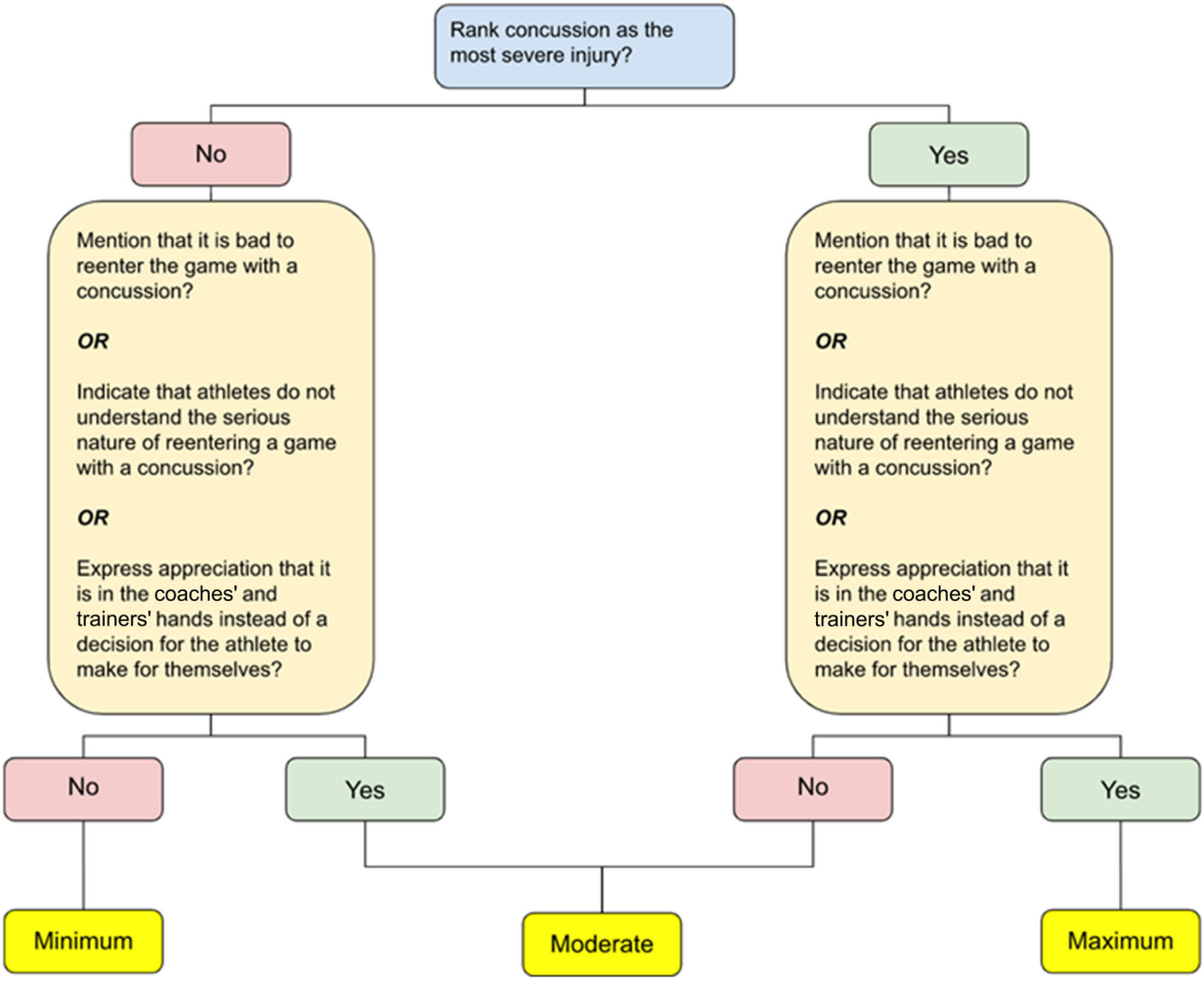

FIGURE 1 Coding criteria.

Coding criteria

Our coding criteria for concussion severity is described in Figure 1 above.

Our coding criteria for brain terminology was as follows:

- Participants were categorized as "Yes" if:
  - There was explicit mention of concussion involving the use of the word "brain" in the context of the open-ended questions.
- Otherwise, participants were categorized as "No."

Statistical analysis

Chi-square was used to assess the relationship between the use of brain terminology (yes/no) and perceived concussion severity (maximum/moderate/minimum) for our primary analysis, as well as for secondary analyses by participant role. For analyses with fewer than five expected participants in any cell, Fisher's exact test was used. To better quantify the relationship between perceived concussion severity and terminology for all significant relationships, odds ratios (OR) and 95% confidence intervals (CI) were also calculated, with "severe" concussion severity as the default level. Analyses were conducted using R version 4.1.2.

RESULTS

The 94 participants included 45 coaches, 18 parents, 22 students, 16 educators, and 1 athletic director and represented 19 U.S. states. Some participants were counted in multiple categories due to overlapping identities (e.g., athletic director and coach; educator and parent). The participants were 55% male ($n = 52$) and 45% female ($n = 42$), 31% were familiar with the researcher prior to research ($n = 29$), and 69% had no

19341563, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/pmrj.13256 by Ohio State University Ohio Sta, Wiley Online Library on [01/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

TABLE 1 Perceived severity of concussions and use of brain phrase.

Participant type n(%)	Severity	No brain phrase	Used brain phrase	*p* value
All participants 94 (100%)	Minimum	18 (95%)	1 (5%)	$p < .001$
	Moderate	33 (77%)	10 (23%)	
	Maximum	14 (44%)	18 (56%)	
Coaches 45 (48%)	Minimum	13 (93%)	1 (7%)	$p = 0.002$
	Moderate	10 (83%)	2 (17%)	
	Maximum	8 (42%)	11 (58%)	
Parents 18 (19%)	Minimum	1 (100%)	0	$p = 0.45$
	Moderate	6 (75%)	2 (25%)	
	Maximum	4 (44%)	5 (56%)	
Educators 16 (17%)	Minimum	2 (100%)	0	$p = 0.16$
	Moderate	9 (90%)	1 (10%)	
	Maximum	2 (50%)	2 (50%)	
Athletes 22 (23%)	Minimum	3 (100%)	0	$p = 0.53$
	Moderate	10 (67%)	5 (33%)	
	Maximum	2 (50%)	2 (50%)	
Athletic director 1 (1%)	Minimum	0	0	n/a
	Moderate	0	0	
	Maximum	1 (100%)	0	

Note: Study consists of 94 unique participants. Some participants were counted in multiple categories due to overlapping identities (e.g., Athletic Director and Coach; Educator and Parent).

prior relationship with the researchers ($n = 65$). Across all participants, 29 used brain terminology to describe concussions, whereas 65 did not. Concussion severity was rated as minimum by 19 participants, moderate by 43 participants, and severe by 32 participants.

Quantitative results

Brain phrases were associated with perceived concussion severity ($p < .001$), and those who described concussions with maximum severity terminology had higher odds of using brain phrases than those who described concussions as having minimum (OR = 0.05, 95% CI = 0.00–0.30, $p < .001$) or moderate severity (OR = 0.24, 95% CI = 0.09–0.65, $p = .003$). In secondary analyses, this significant relationship between the use of brain terminology and perceived concussion severity was preserved for coaches ($p = .002$), but not for other groups (all $p > .05$). Coaches who described concussions as severe were more likely to use brain terminology than coaches who described concussions as minimum severity (OR = 0.06, 95% CI = 0.00–0.41, $p = .003$) or moderate severity (OR = 0.14, 95% CI = 0.02–0.76, $p = .026$). We had 80% power to detect medium-large effects (w = 0.73) for parents, large effects (w = 0.89) for educators, medium effects (w = 0.66) for athletes, and very large effects (w = 3.31) for coaches.

Qualitative findings

Interviewees showed a wide range of beliefs about concussions and their severity. Responses differed by the role of the participant (coach, parent, athlete, educator). A full list of quotes with the assigned severity code can be found in Data S2. Some participants met the criteria of the maximum severity category and included brain phrases, as seen with these coaches:

> "*The way I look at concussions is you can replace a broken bone in your body. You can heal a sprained ankle, sprained thumb, whatever the case may be. But when it comes to your head and your brain, that's something that you cannot replace*."

> "*We're talking about the brain here, so that's something that can have long-term effects for the rest of their life… you know, you can cut off your leg and survive, but you're not going to cut off your head and survive*."

Parents unsurprisingly reflected on their personal experiences dealing with concussions that their children had suffered. One such parent who met the criteria for the maximum severity category recalled the consequences of concussions suffered by both her son and daughter:

19341563, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/pmrj.13256 by Ohio State University Ohio Sta, Wiley Online Library on [01/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

> "*My son has had two concussions and it really affected his schoolwork. It was very scary … the first one was terrible. You'd be talking to him and he'd just fall asleep almost. And my daughter had a concussion too, and it was really hard those weeks where they really just needed to let their brains heal. And then it's scary to think that they could get another one and the consequences of that*."

Athletes had their own reasoning for seeing concussions as severe. Whereas some explicitly stated that it was because of concussions they had suffered in the past, others simply cited the importance of the brain:

> "*For the most severe option, I chose concussion just because the brain is very important to functionality and day-to-day living more than like an arm or an ankle*."

Some participants, like the following educator, ranked concussion as the most severe injury and met two of the other criteria without mentioning the word "brain." In this case, the participant expressed both the belief that the athletes did not understand negative consequences associated with premature return to play and mentioned that it would be unwise for athletes to return to play prematurely:

> "*I know that concussions can impact you well into your future … I would hope that adults would encourage against [going back into a game with a concussion]. Like if I think if the kid had the choice and it was solely up to the kid, do you want to go back in this game or not? I think they could potentially not see [the concussion] as severe, because I think if I was ranking those injuries from most painful to least painful, I don't think concussions would be the most painful or most obvious. So, I think the kids themselves would choose [to go back into the game] but if they had any sort of coach or adult I hope somebody would tell them not to*."

One athlete exemplifies the 19 participants who expressed concussions have minimum severity. This athlete ranked a torn ACL as more severe than a concussion, and in response to the question "if an athlete were to get injured and find out they had a concussion, do you think they would choose to go back into the game if presented with the choice?" the participant said the following:

> "*I would not put myself back in the game just because I'm not a big fan of injuries. And when I get hurt, I kind of know it. And I just try and sideline myself … but for somebody else, it really depends on the person. I would say some people who really want to be playing and who just don't really feel pain as much, would stick in the game, but like someone who's more vulnerable to an injury and realized the consequences would probably withdraw themselves. So I would say it really depends on the person*."

Some coaches expressed that concussions had minimum severity. In one instance, a coach suggested that most athletes would want to remain playing after a concussion:

> "*Clearly the real competitive ones would [go back into the game], if a kid didn't play much and he had an opportunity to play, and he was having a good game for whatever reason he would want to go back in as well. Very few of them are going to say 'Uh, this could really, I could really get hurt by this concussion and I better not suffer another fall or get hit in the head or something*.'"

Many participants expressed that concussions had moderate severity. Some of these participants ranked concussion as the most severe injury but did not meet any of the other criteria, as expressed by a coach in response to a question whether or not athletes would choose to go back into the game with a concussion:

> "*Probably. It would depend on the severity and what sort of symptoms they're experiencing at that point in time, but for relatively minor ones, absolutely. They want to go back in*."

In elaborating, this coach did not express appreciation that adult stakeholders were responsible for return to play decision-making, did not indicate that athletes did not understand negative consequences associated with premature return to play, and did not express his opinion that it would be unwise to go back into the game prematurely with a concussion.

DISCUSSION

This is the first study to our knowledge that has investigated the relationship between terminology use and

19341563, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/pmrj.13256 by Ohio State University Ohio Sta, Wiley Online Library on [01/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

perceived severity of concussions. The results suggest that participants who perceive concussions as a severe injury were more likely to use the word "brain" when speaking about concussions. When broken down and analyzed by group, this relationship was statistically significant among coaches interviewed but not for athletes, parents, educators, or athletic directors. Given the importance and ubiquity of language in conveying experience, there are key takeaways from this work.

First, developing the use of appropriate and accurate language to describe concussion is critical. This mandate is substantiated not only by this study, but by prior work demonstrating the importance of defining concussion in research and in measuring differences in concussion recognition when using different phrases.[23] This recommendation is also reflective of endeavors that have sought to improve the language used by media broadcasters in describing concussion when it occurs in competition.[25,26] Broadly, if language can alter individuals' perceptions of the nature, likelihood, or severity of concussion, it is critical that precise and accurate language is used scientifically, educationally, and in lay media to avoid over- or underemphasizing the nature of the injury. A major takeaway from the present study is the necessary emphasis that *a concussion represents a brain injury*.

There are several plausible means by which language may influence athlete concussion care-seeking behavior. First, this study suggests that language is related to the perceived severity of concussion; perceived concussion severity has been found in prior research to be associated with a greater intent to report concussions.[18,19] Alternatively, the language used could influence perceived social norms. If, for example, a coach or teammate uses language that minimizes concussion severity, an athlete may interpret this as a signal that the environment norms are unsupportive of concussion reporting. Conversely, if an athlete's coach or teammates conveys that concussions should be taken seriously, the athlete may take this language as supportive of concussion care-seeking behaviors. These perceived social norms have frequently been demonstrated to be related to behavioral outcomes in the context of concussion and other processes.[20–22]

Investigating the language used surrounding concussions provides an important window into concussion decision-making, thereby yielding potential opportunities for educational interventions. An improved understanding of the role of language can be used to inform athlete and coach educational programs for improving health outcomes of concussed athletes. These findings are consequential in the context of previous work suggesting that the vocabulary used to describe and define concussions also has an impact on self-reporting.[23,24]

A notable finding from our study was that coaches were the only group with a statistically significant relationship between the use of brain terminology and perceived concussion severity. It is possible that the use of brain terminology might reflect the fact that medical knowledge has a greater effect on coaches' beliefs as compared with noncoaches, which would support the use of mandated medical education programs. Alternatively, it is possible that this study was inadequately powered to detect differences in the noncoach subgroups.

Strengths of the present study include the open-ended response design facilitated by the interviewer that saw each participant answering the same questions while providing room to respond freely. Access to audio recordings of each participant interview allowed for a second coder and therefore added to the reliability of our coding methods. Interviewees were diverse in their group belonging and represented 19 different U.S. states.

Limitations of the analysis include potential sample bias because participants were predominantly contacted via cold emails and LinkedIn messages. It is possible that outside factors such as preexisting familiarity with the interviewer influenced responses. As with nearly all qualitative work, the target was not a large generalizable sample; thus, it may be the case that this trend would be different in another population. Semistructured interviews, although advantageous in some regards, also allowed some participants to not be elaborative in their language and may have hindered the ability to properly gauge their beliefs about concussions. This happened when participants were asked to explain why they ranked the injuries in the order they did, to which some participants had discussed each of the injuries in detail and their rationale for relative ranking of injuries, whereas other participants simply noted an experience with one of the injuries with limited discussion of the other injuries. As a result, some participants provided more insight into their concussion views compared to other participants. Furthermore, the interviewees might have ranked the "severity" of injury according to differing criteria given the open-ended nature of the question. Severity could be taken to mean the pain of an injury, the duration of time an athlete would be forced to sit out from athletic activity, or how an injury could affect an athlete long term.

CONCLUSION

Individuals who perceive concussions as more severe also tend to describe concussion specifically as brain terminology, highlighting the importance of precise medical language in describing concussion. These findings can inform the way that concussions are discussed and the language used in concussion education administered to coaches, athletes, and parents, as well as more broadly in the lay media. Describing concussions using

19341563, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/pmrj.13256 by Ohio State University Ohio Sta, Wiley Online Library on [01/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

medical terminology and using brain phrases could provide benefit for athlete concussion reporting.

ACKNOWLEDGMENTS

The work of Ms Katherine Schneider was supported by a grant from the Haas Center for Public Service Undergraduate Fellowships Program at Stanford University. The funding agency was not involved in study oversight, data acquisition, data analysis, or reporting. We are grateful to our dedicated research assistants who include Rosa Hamalainen, Christine Lynn Chen, Anafia Thomas, Courtney Gao, and Maya Rae Lorey. We appreciate the time and insights of our study participants. Finally, our heartfelt thanks to Coach Dick Gould for his tireless leadership and commitment to bring rigorous concussion education to the global sports community.

DISCLOSURES

Piya Sorcar: Education Advisory Council for USA Football. Daniel Daneshvar: Medical Advisor and Options Holder, StataDx. Medical expert testimony related to SCI and TBI. Associate Editor, PM&R Journal. Grant recipient from National Institute of Neurological Disorders and Stroke and Foundation for PM&R (neither of which supported the work described in the present manuscript). Unpaid leadership positions include Chair of ALS Task Force at ACRM, Executive Advisory Board at Texas Law Enforcement Peer Network, Research Committee at AAP, Director of Institute for Brain Research and Innovation at TeachAids, and Face the Fight Scientific Advisory Committee.

ORCID

Amy K. Bugwadia https://orcid.org/0000-0002-3509-3556
Roy D. Pea https://orcid.org/0000-0001-6301-3536
Daniel H. Daneshvar https://orcid.org/0000-0003-3691-9513
Christine M. Baugh https://orcid.org/0000-0002-9340-6338

REFERENCES

1. Bryan MA, Rowhani-Rahbar A, Comstock RD, Rivara F, on behalf of the Seattle Sports Concussion Research Collaborative. Sports- and recreation-related concussions in US youth. *Pediatrics*. 2016;138(1):e20154635. doi:10.1542/peds.2015-4635
2. Daneshvar DH, Nowinski CJ, McKee A, Cantu RC. The epidemiology of sport-related concussion. *Clin Sports Med*. 2011;30(1):1-17. doi:10.1016/j.csm.2010.08.006
3. McCrea M, Hammeke T, Olsen G, Leo P, Guskiewicz K. Unreported concussion in high school football players: implications for prevention. *Clin J Sport Med*. 2004;14(1):13-17. doi:10.1097/00042752-200401000-00003
4. Register-Mihalik JK, Linnan LA, Marshall SW, Valovich McLeod TC, Mueller FO, Guskiewicz KM. Using theory to understand high school aged athletes' intentions to report sport-related concussion: implications for concussion education initiatives. *Brain Inj*. 2013;27(7–8):878-886. doi:10.3109/02699052.2013.775508
5. Wallace J, Covassin T, Nogle S, Gould D, Kovan J. Knowledge of concussion and reporting behaviors in high school athletes with or without access to an athletic trainer. *J Athl Train*. 2017;52(3):228-235. doi:10.4085/1062-6050-52.1.07
6. Moore IS, Bitchell CL, Vicary D, Rafferty J, Robson BC, Mathema P. Concussion increases within-player injury risk in male professional rugby union. *Br J Sports Med*. 2023;57(7):395-400. doi:10.1136/bjsports-2021-105238
7. Nordström A, Nordström P, Ekstrand J. Sports-related concussion increases the risk of subsequent injury by about 50% in elite male football players. *Br J Sports Med*. 2014;48(19):1447-1450. doi:10.1136/bjsports-2013-093406
8. Daneshvar DH, Riley DO, Nowinski CJ, McKee AC, Stern RA, Cantu RC. Long term consequences: effects on normal development profile after concussion. *Phys Med Rehabil Clin N Am*. 2011;22(4):683-700. doi:10.1016/j.pmr.2011.08.009
9. Karlin AM. Concussion in the pediatric and adolescent population: "different population, different concerns". *PM R*. 2011;3-(10 Suppl 2):S369-S379. doi:10.1016/j.pmrj.2011.07.015
10. McKee AC, Cantu RC, Nowinski CJ, et al. Chronic traumatic encephalopathy in athletes: progressive tauopathy after repetitive head injury. *J Neuropathol Exp Neurol*. 2009;68(7):709-735. doi:10.1097/nen.0b013e3181a9d503
11. Collins MW, Lovell MR, Iverson GL, Cantu RC, Maroon JC, Field M. Cumulative effects of concussion in high school athletes. *Neurosurgery*. 2002;51(5):1175-1181. doi:10.1097/00006123-200211000-00011
12. Moser RS, Schatz P, Jordan BD. Prolonged effects of concussion in high school athletes. *Neurosurgery*. 2005;57(2):300-306. doi:10.1227/01.neu.0000166663.98616.e4
13. Howell DR, Kriz P, Mannix RC, Kirchberg T, Master CL, Meehan WP. Concussion symptom profiles among child, adolescent, and Young adult athletes. *Clin J Sport Med*. 2019;29(5):391-397. doi:10.1097/JSM.0000000000000629
14. McPherson AL, Nagai T, Webster KE, Hewett TE. Musculoskeletal injury risk after sport-related concussion: a systematic review and meta-analysis. *Am J Sports Med*. 2019;47(7):1754-1762. doi:10.1177/0363546518785901
15. Bey T, Ostick B. Second impact syndrome. *West J Emerg Med*. 2009;10(1):6-10.
16. Harmon KG, Drezner JA, Gammons M, et al. American medical society for sports medicine position statement: concussion in sport. *Br J Sports Med*. 2013;47(1):15-26. doi:10.1136/bjsports-2012-091941
17. Kurowski BG, Pomerantz WJ, Schaiper C, Ho M, Gittelman MA. Impact of preseason concussion education on knowledge, attitudes, and behaviors of high school athletes. *J Trauma Acute Care Surg*. 2015;79:S21-S28. doi:10.1097/TA.0000000000000675
18. Register-Mihalik JK, Guskiewicz KM, McLeod TCV, Linnan LA, Mueller FO, Marshall SW. Knowledge, attitude, and concussion-reporting behaviors among high school athletes: a preliminary study. *J Athl Train*. 2013;48(5):645-653. doi:10.4085/1062-6050-48.3.20
19. Glasman LR, Albarracín D. Forming attitudes that predict future behavior: a meta-analysis of the attitude-behavior relation. *Psychol Bull*. 2006;132(5):778-822.
20. Daneshvar DH, Baugh CM, Yutsis M, et al. Athlete enjoyment of prior education moderates change in concussion-reporting intention after interactive education. *Inquiry*. 2021;58:469580211022641. doi:10.1177/00469580211022641
21. Daneshvar DH, Yutsis M, Baugh CM, et al. Evaluating the effect of concussion-education programs on intent to report concussion in high school football. *J Athl Train*. 2021;56(11):1197-1208.
22. Kroshus E, Baugh CM, Daneshvar DH, Viswanath K. Understanding concussion reporting using a model based on the theory of planned behavior. *J Adolesc Health*. 2014;54(3):269-274.e2. doi:10.1016/j.jadohealth.2013.11.011

19341563, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/pmrj.13256 by Ohio State University Ohio Sta, Wiley Online Library on [01/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License

23. Robbins CA, Daneshvar DH, Picano JD, et al. Self-reported concussion history: impact of providing a definition of concussion. *Open Access J Sports Med*. 2014;5:99. doi:10.2147/OAJSM.S58005
24. Miyashita TL, Diakogeorgiou E, Hellstrom B, Kuchwara N, Tafoya E, Young L. High school athletes' perceptions of concussion. *Orthop J Sports Med*. 2014;2(11):2325967114554549. doi:10.1177/2325967114554549
25. Media Project | Concussion Legacy Foundation. (n.d.). Retrieved August 20, 2022, from https://concussionfoundation.org/programs/media-project
26. Ahmed OH, Hall EE. "It was only a mild concussion": exploring the description of sports concussion in online news articles. *Phys Ther Sport*. 2017;23:7-13. doi:10.1016/j.ptsp.2016.07.003

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Knowles P, Schneider K, Bugwadia AK, et al. The importance of language in describing concussions: A qualitative analysis. *PM&R*. 2024;1-8. doi:10.1002/pmrj.13256

19341563, 0, Downloaded from https://onlinelibrary.wiley.com/doi/10.1002/pmrj.13256 by Ohio State University Ohio Sta, Wiley Online Library on [01/10/2024]. See the Terms and Conditions (https://onlinelibrary.wiley.com/terms-and-conditions) on Wiley Online Library for rules of use; OA articles are governed by the applicable Creative Commons License